

CMOS Inverter Layout Project Report

Schematic, Layout, DRC and LVS Verification in Cadence Virtuoso

Item	Details
Project	CMOS inverter custom layout
Software	Cadence Virtuoso Layout Suite L
Verification tool	Assura DRC and Assura LVS
Technology / PDK	gpdk090
Devices used	pmos1v and nmos1v
Library / Cell	Anik / CMOS inverter
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Note: This report is intentionally written in simple language so that a reader with basic CMOS knowledge can understand the full project flow.

1. Project Summary

This project documents the complete design and verification flow of a CMOS inverter layout in Cadence Virtuoso. The work started from a schematic using one PMOS and one NMOS transistor. After that, a custom layout was made manually. The layout was then checked using DRC and LVS. Finally, both checks passed successfully: no DRC errors were found, and the schematic and layout matched in LVS.

1.1 Main objective

- Create a CMOS inverter schematic using pmos1v and nmos1v devices.
- Create the physical layout of the inverter in Cadence Virtuoso.
- Connect source, drain, gate and body terminals correctly.
- Add layout pins A, Y, VDD and VSS.
- Run DRC to check physical design-rule correctness.
- Run LVS to confirm that the layout matches the schematic.

1.2 Final result

Item	Details
DRC result	Passed. No DRC errors found.
LVS result	Passed. Schematic and Layout Match.
Net mismatches	0
Device mismatches	0
Pin mismatches	0
Parameter mismatches	0

2. Basic CMOS Inverter Theory

A CMOS inverter is one of the most basic digital CMOS circuits. It uses one PMOS transistor as the pull-up device and one NMOS transistor as the pull-down device. The input is connected to both gates. The output is taken from the common drain connection of the PMOS and NMOS.

Item	Details
Input A = 0	PMOS turns ON, NMOS turns OFF, output Y goes to VDD.
Input A = 1	PMOS turns OFF, NMOS turns ON, output Y goes to VSS.

In the layout, the PMOS source and body must be connected to VDD. The NMOS source and body must be connected to VSS. These body connections are important because they prevent unwanted body bias problems and also help LVS recognize the correct circuit.

3. Schematic Design

The schematic was created first. This is important because LVS later compares the final layout with this schematic. The schematic cell was created as Anik / CMOS_inverter / schematic.

3.1 Schematic devices and sizes

Item	Details
PMOS device	pmos1v
NMOS device	nmos1v
PMOS width	480 nm
NMOS width	240 nm
Channel length	100 nm for both devices
Multiplier	m = 1

3.2 Schematic connections

Item	Details
PMOS source	VDD
PMOS body / bulk	VDD
PMOS gate	A
PMOS drain	Y
NMOS drain	Y
NMOS gate	A
NMOS source	VSS
NMOS body / bulk	VSS

Figure 1: Schematic placeholder

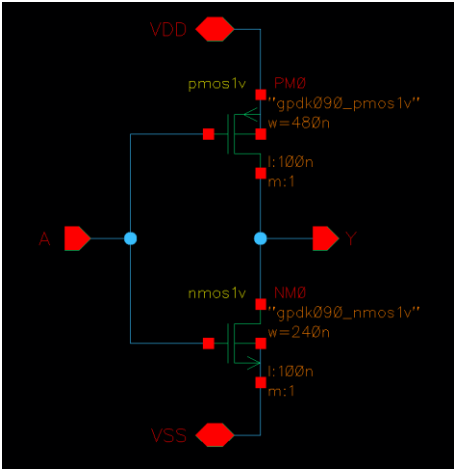
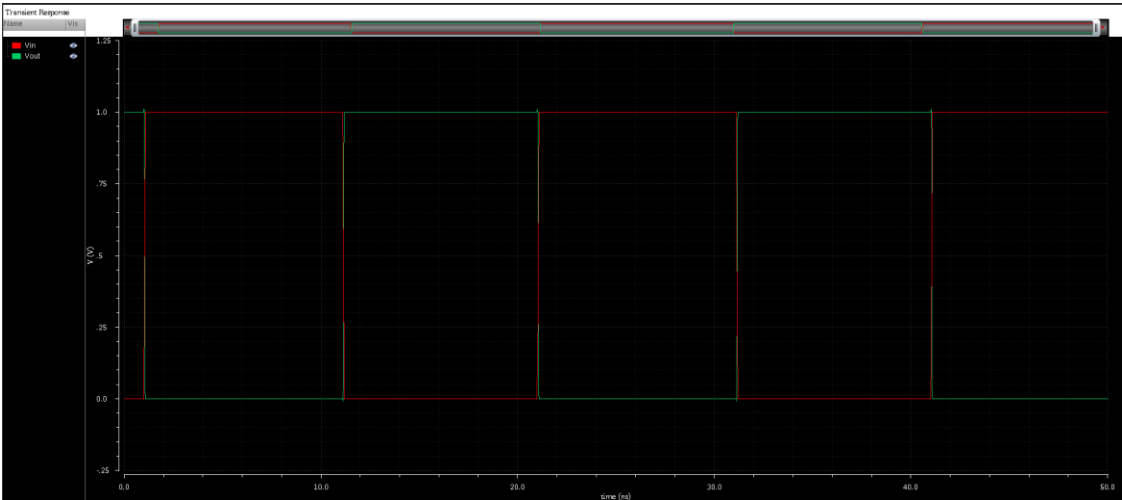


Figure 2: Simulation / graph placeholder



4. Layout Design Flow

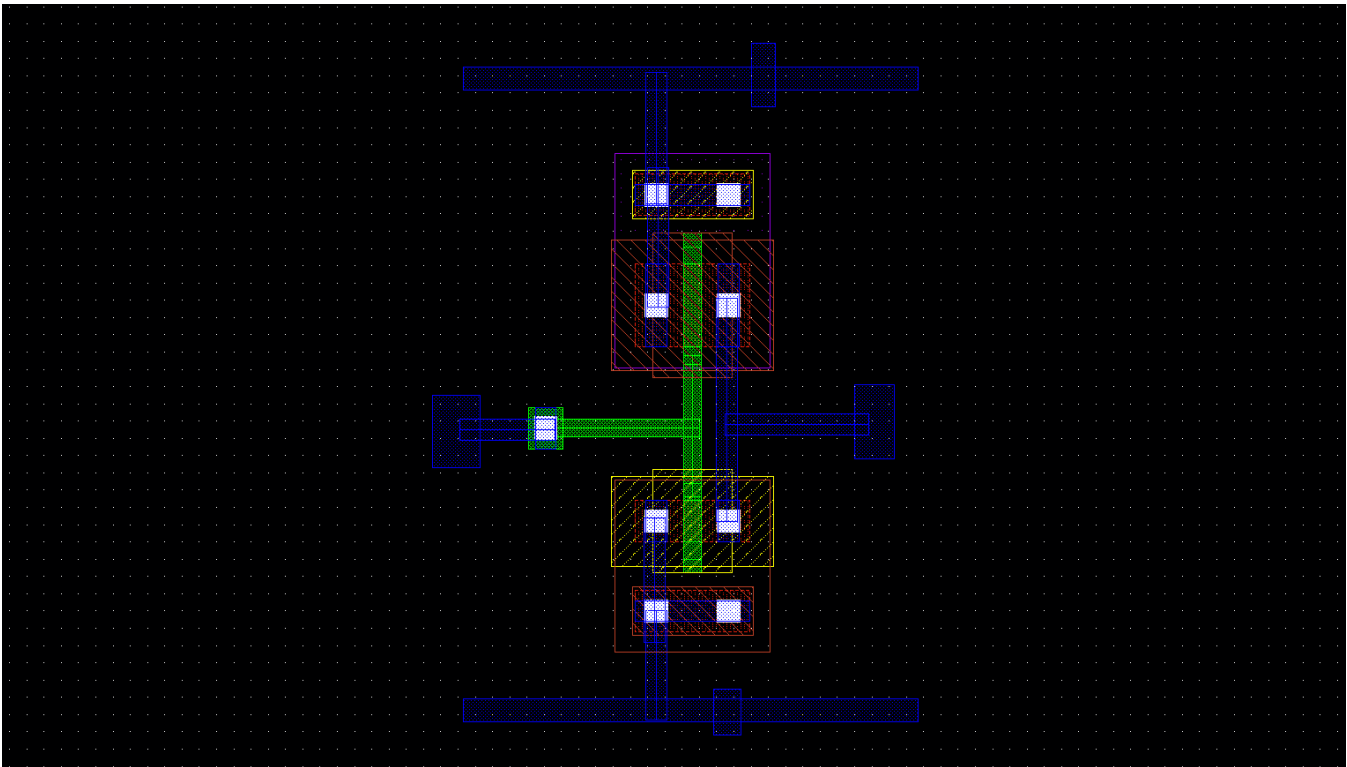
After the schematic was completed and checked, a layout view was created under the same cell name: Anik / CMOS_inverter / layout. The same cell name was used so that Assura LVS could compare schematic and layout correctly.

4.1 Device placement

The PMOS layout instance was placed above the NMOS layout instance. This is a standard inverter layout style because the PMOS connects to the upper VDD rail and the NMOS connects to the lower VSS rail. The green poly gates of both devices were aligned vertically so the input gate connection could be made easily.

Item	Details
Top device	PMOS
Bottom device	NMOS
Top rail	VDD
Bottom rail	VSS
Left side diffusion	Used as source side
Right side diffusion	Used as drain/output side

Figure 3: Final layout placeholder



5. Layout Connections

The physical connections were made using Metall, poly and proper contacts. Metall was used for the supply rails, output wire and input access wire. Poly was used to connect the gates of the PMOS and NMOS.

Item	Details
VDD rail	A horizontal Metall rail was drawn above the PMOS.
VSS rail	A horizontal Metall rail was drawn below the NMOS.
PMOS source	Connected to VDD using Metall.
NMOS source	Connected to VSS using Metall.
Output Y	PMOS drain and NMOS drain were connected together and extended to the right.
Input A	PMOS gate and NMOS gate were connected using poly, then connected to Metall through M1 POv contact.

5.1 Body connections

The body connections were an important part of this layout. At first, the body tap was not easy to find manually from the instance list. Later, the PMOS and NMOS properties were used to select a detached body type. This created separate body contacts. The PMOS body contact was connected to VDD and the NMOS body contact was connected to VSS.

Item	Details
PMOS body	Connected to VDD using the detached body contact.
NMOS body	Connected to VSS using the detached body contact.
Reason	Correct body connection is needed for proper transistor operation and LVS matching.

5.2 Layout pins

Four layout pins were added using Create Shape Pin. The pin names were kept exactly the same as the schematic pin names. This was necessary because mismatched pin names can cause LVS errors.

Item	Details
A	Input pin, placed on the left Metall input wire.
Y	Output pin, placed on the right Metall output wire.
VDD	Power pin, placed on the top Metall rail.
VSS	Ground pin, placed on the bottom Metall rail.

6. Problems Faced and How They Were Solved

Several small issues were faced during the project. These issues are common when learning custom layout for the first time. The table below records the problem and the solution used.

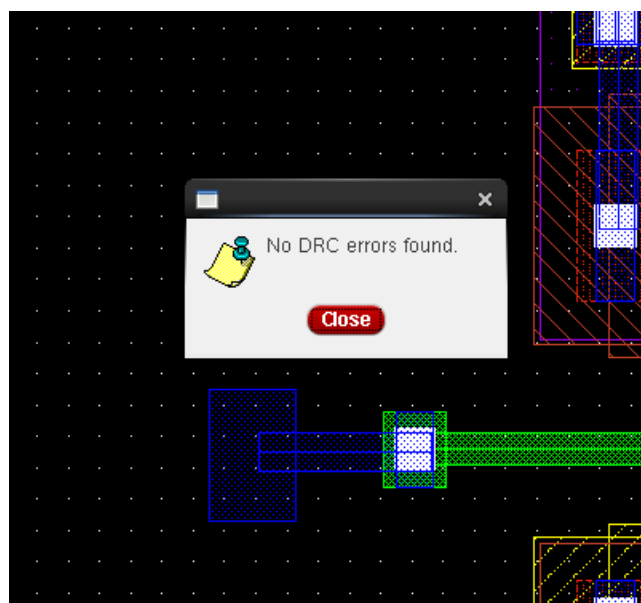
Problem	Reason	Solution
Only pmos1v and nmos1v text was visible in layout.	Hierarchy was not expanded.	Pressed Shift + F to display the actual transistor layers.
Rectangle drawing did not work properly at first.	Cadence rectangle command needed two clicks, not click-drag-release.	Used two-click method and also used path drawing for Metall rails.
Right-side schematic connection looked shorted at first.	VDD, Y and VSS could accidentally become one continuous net.	Separated the nets so VDD, Y and VSS were independent.
Poly gate connection was initially wrong.	The poly path was touching the white source/drain contact area.	Deleted the wrong poly loop and connected only the center green gate regions.
Poly-to-Metall contact name was confusing.	The contact was named M1_POv instead of normal V1.	Used M1_POv for the input gate contact because it connects poly to Metall.
N-well tap was hard to find in the instance list.	The PDK did not show an obvious ntap cell name.	Used the transistor body type property and selected detached body contact.
Assura DRC technology was undefined.	The DRC form did not automatically select the PDK technology at first.	Selected the correct gpd090 technology/rule setup before running DRC.

7. DRC Verification

DRC means Design Rule Check. It checks whether the drawn layout follows the physical rules of the technology, such as minimum width, spacing, enclosure and contact rules. After correcting the layout and body contact issues, DRC was run using Assura.

Item	Details
Tool	Assura DRC
Layout source	DFII
Library	Anik
Cell	CMOS_inverter
View	layout
Result	No DRC errors found.

Figure 4: DRC result placeholder

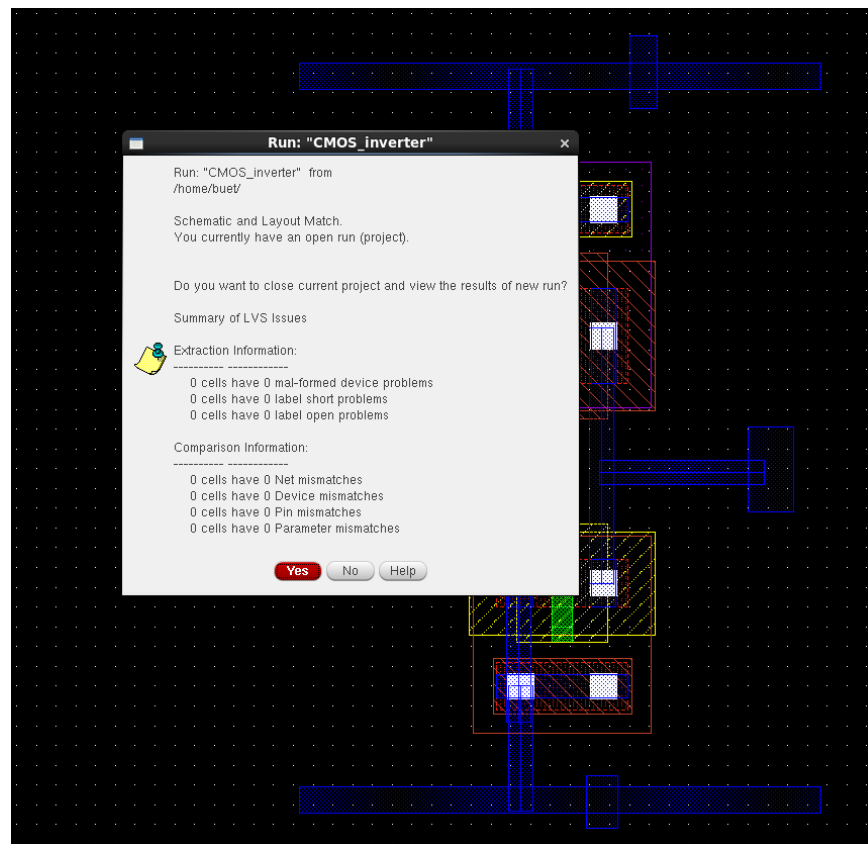


8. LVS Verification

LVS means Layout Versus Schematic. It checks whether the layout contains the same circuit as the schematic. Even if a layout passes DRC, it can still be electrically wrong. Therefore LVS is very important.

Item	Details
Tool	Assura LVS
Schematic view	Anik / CMOS_inverter / schematic
Layout view	Anik / CMOS_inverter / layout
Result	Schematic and Layout Match.
Net mismatches	0
Device mismatches	0
Pin mismatches	0
Parameter mismatches	0

Figure 5: LVS result placeholder



9. Final Project Checklist

- CMOS inverter schematic created successfully.
- pmoslv and nmoslv devices used correctly.
- PMOS W = 480 nm and NMOS W = 240 nm used in the design.
- PMOS placed at top and NMOS placed at bottom in layout.
- VDD and VSS Metall rails created.
- Sources connected to supply rails correctly.
- Drains connected together as output Y.
- Gates connected together as input A.
- Detached body contacts added and connected correctly.
- Pins A, Y, VDD and VSS added with matching names.

- Assura DRC completed with zero errors.
- Assura LVS completed with schematic and layout match.

10. Conclusion

The CMOS inverter layout project was completed successfully. The project started from a simple schematic and ended with a verified custom layout. The main learning points were MOSFET terminal connection, Metal routing, poly gate connection, body tap connection, layout pins, DRC and LVS. The final layout passed DRC with no errors and passed LVS with schematic and layout match. This confirms that the layout is both physically valid and electrically correct.